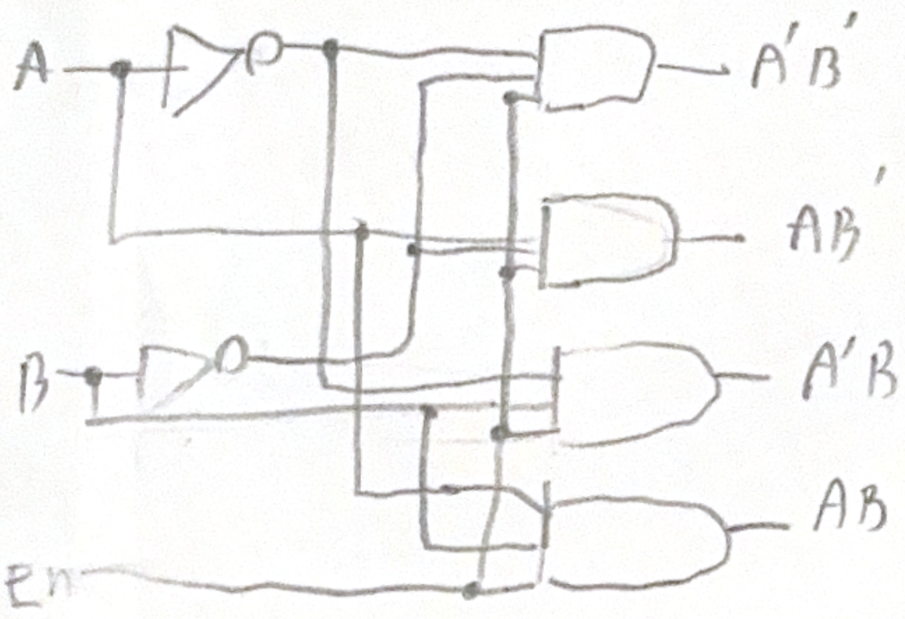
Exercise: Self-Checking Circuit

ID no. 6733701021 Name Thanakrit Numnuan

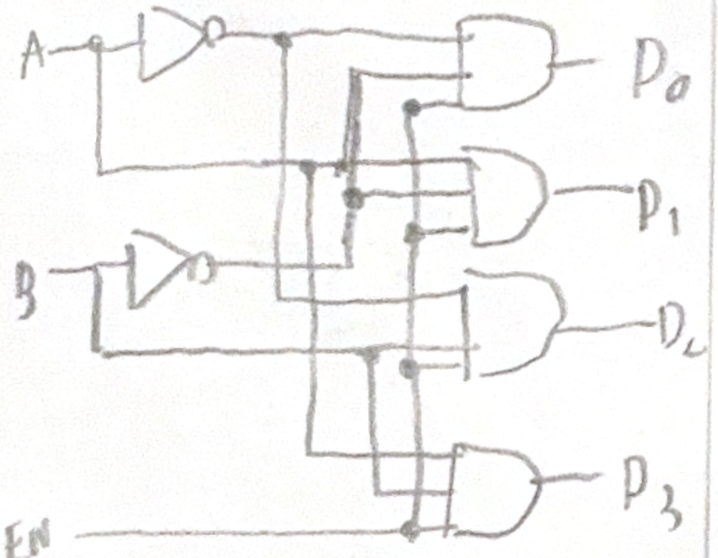
1. Design a Self-Checking 2:4 Demultiplexer with enable



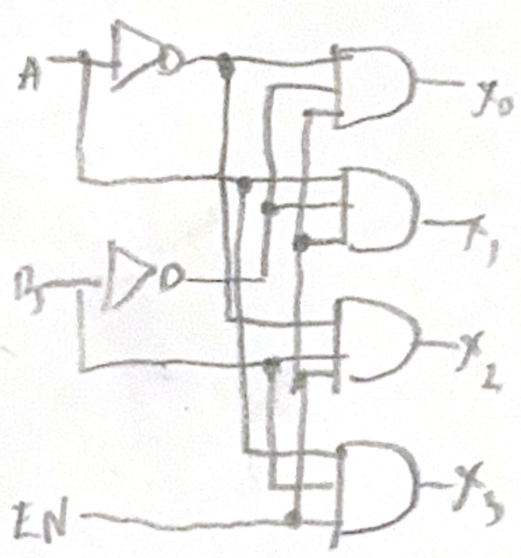
E	A	B	D_0	D_1	D_2	D_3
0	X	X	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

$D_0 = EN \cdot A' \cdot B'$ $D_1 = EN \cdot A' \cdot B$ $D_2 = EN \cdot A \cdot A'$ $D_3 = EN \cdot A \cdot B$

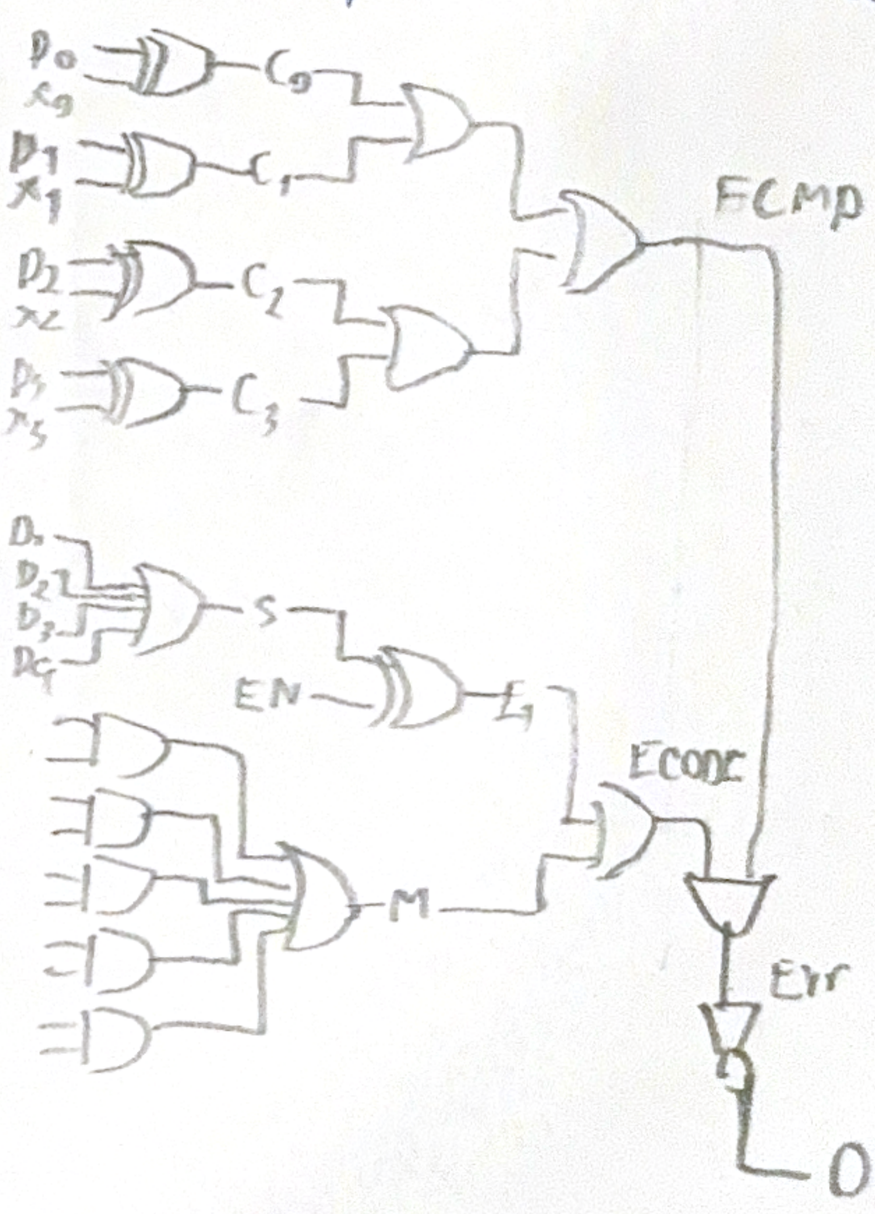
original



Expected output



comparator + One hot checker



$C_i = D_i \oplus X_i$

$ECMP = C_0 + C_1 + C_2 + C_3$

$S = D_0 + D_1 + D_2 + D_3$

$M = D_0 D_1 + D_0 D_2 + D_0 D_3 + D_1 D_2 + D_1 D_3 + D_2 D_3$

$ECODE = (EN \oplus S) + M$

$ERR = (EN \oplus S) + M$

$OK = ERR'$